Ordinary Differential Equations

Lecture Notes

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Lecture 4: Eigenvalue Methods and Diagonalization

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1 Introduction to Eigenvalue Methods

Eigenvalue methods provide the most powerful and systematic approach to solving linear systems of differential equations. These methods not only yield explicit solutions but also reveal the fundamental structure underlying the system's behavior. The eigenvalue-eigenvector decomposition transforms complex coupled systems into collections of independent, simpler equations.

The central idea is to find special directions in the state space—the eigenvector directions—along which the system evolves in the simplest possible way. Along these directions, the system behaves like a one-dimensional equation with exponential solutions.

For the linear system $\frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x}$, we seek solutions of the form:

$$\mathbf{x}(t) = \mathbf{v}e^{\lambda t} \tag{1}$$

where $\mathbf{v}$ is a constant vector and λ is a constant scalar. Substituting into the differential equation:

$$\lambda \mathbf{v}e^{\lambda t} = \mathbf{A}\mathbf{v}e^{\lambda t} \tag{2}$$

This leads to the eigenvalue problem:

$$\mathbf{A}\mathbf{v} = \lambda \mathbf{v} \tag{3}$$

The scalars λ are eigenvalues and the corresponding vectors $\mathbf{v}$ are eigenvectors of matrix $\mathbf{A}$.

2 The Eigenvalue Problem

Eigenvalues and Eigenvectors For an $n \times n$ matrix $\mathbf{A}$, a scalar λ is an eigenvalue and a nonzero vector $\mathbf{v}$ is a corresponding eigenvector if:

$$\mathbf{A}\mathbf{v} = \lambda \mathbf{v} \tag{4}$$

Equivalently, $(\mathbf{A} - \lambda \mathbf{I})\mathbf{v} = \mathbf{0}$, which has nontrivial solutions if and only if:

$$\det(\mathbf{A} - \lambda \mathbf{I}) = 0 \tag{5}$$

This equation is called the characteristic equation, and its left side is the characteristic polynomial.

2.1 Computing Eigenvalues

The characteristic polynomial of an $n \times n$ matrix is a polynomial of degree n:

$$p(\lambda) = \det(\mathbf{A} - \lambda\mathbf{I}) = (-1)^n\lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_1\lambda + c_0 \tag{6}$$

The coefficients are related to the matrix elements through: - $c_{n-1} = (-1)^{n-1}\mathrm{tr}(\mathbf{A})$ (trace) - $c_0 = \det(\mathbf{A})$ (determinant)

For a 2×2 matrix $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$:

$$p(\lambda) = \lambda^2 - (a+d)\lambda + (ad - bc) = \lambda^2 - \mathrm{tr}(\mathbf{A})\lambda + \det(\mathbf{A}) \tag{7}$$

The eigenvalues are:

$$\lambda_{1,2} = \frac{\mathrm{tr}(\mathbf{A}) \pm \sqrt{(\mathrm{tr}(\mathbf{A}))^2 - 4\det(\mathbf{A})}}{2} \tag{8}$$

2.2 Computing Eigenvectors

Once eigenvalues are found, eigenvectors are computed by solving:

$$(\mathbf{A} - \lambda_i\mathbf{I})\mathbf{v}_i = \mathbf{0} \tag{9}$$

This is a homogeneous linear system. The eigenvector is determined up to a scalar multiple, so we typically normalize it or choose a convenient scaling.

Example 2.1: Complete Eigenvalue Analysis

onsider the matrix:

$$\mathbf{A} = \begin{pmatrix} 3 & 1 \\ 2 & 2 \end{pmatrix} \tag{10}$$

Step 1: Find eigenvalues

$$\det(\mathbf{A} - \lambda\mathbf{I}) = \det\begin{pmatrix} 3-\lambda & 1 \\ 2 & 2-\lambda \end{pmatrix} = (3-\lambda)(2-\lambda) - 2 = \lambda^2 - 5\lambda + 4 \tag{11}$$

Factoring: $(\lambda - 4)(\lambda - 1) = 0$, so $\lambda_1 = 4$, $\lambda_2 = 1$.
Step 2: Find eigenvectors
For $\lambda_1 = 4$:

$$(\mathbf{A} - 4\mathbf{I})\mathbf{v}_1 = \begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix}\mathbf{v}_1 = \mathbf{0} \tag{12}$$

This gives $-v_1 + v_2 = 0$, so $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

For $\lambda_2 = 1$:

$$(\mathbf{A} - \mathbf{I})\mathbf{v}_2 = \begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix}\mathbf{v}_2 = \mathbf{0} \tag{13}$$

This gives $2v_1 + v_2 = 0$, so $\mathbf{v}_2 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$.

3 Diagonalization

When a matrix has a complete set of linearly independent eigenvectors, it can be diagonalized.

Theorem 3.1: Diagonalization Theorem

n $n \times n$ matrix $\mathbf{A}$ is diagonalizable if and only if it has n linearly independent eigenvectors. In this case:

$$\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1} \tag{14}$$

where $\mathbf{P}$ is the matrix of eigenvectors and $\mathbf{D}$ is the diagonal matrix of eigenvalues:

$$\mathbf{P} = \begin{pmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix} \tag{15}$$

3.1 Solution via Diagonalization

When $\mathbf{A}$ is diagonalizable, the solution to $\frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x}$ can be written as:

$$\mathbf{x}(t) = \mathbf{P}e^{\mathbf{D}t}\mathbf{P}^{-1}\mathbf{x}_0 \tag{16}$$

Since $\mathbf{D}$ is diagonal:

$$e^{\mathbf{D}t} = \begin{pmatrix} e^{\lambda_1 t} & 0 & \cdots & 0 \\ 0 & e^{\lambda_2 t} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & e^{\lambda_n t} \end{pmatrix} \tag{17}$$

This shows that the solution is a linear combination of exponential functions with rates determined by the eigenvalues.

3.2 Modal Coordinates

The transformation $\mathbf{y} = \mathbf{P}^{-1}\mathbf{x}$ introduces modal coordinates. In these coordinates, the system becomes:

$$\frac{d\mathbf{y}}{dt} = \mathbf{D}\mathbf{y} \tag{18}$$

This is a decoupled system:

$$\frac{dy_1}{dt} = \lambda_1 y_1 \tag{19}$$

$$\frac{dy_2}{dt} = \lambda_2 y_2 \tag{20}$$

$$\vdots \tag{21}$$

$$\frac{dy_n}{dt} = \lambda_n y_n \tag{22}$$

Each modal coordinate evolves independently according to $y_i(t) = y_i(0)e^{\lambda_i t}$.

4 Complex Eigenvalues

When the coefficient matrix is real but has complex eigenvalues, they occur in conjugate pairs. This leads to oscillatory solutions.

4.1 Complex Exponentials and Real Solutions

If $\lambda = \alpha + i\beta$ is a complex eigenvalue with eigenvector $\mathbf{v} = \mathbf{u} + i\mathbf{w}$, then:

$$\mathbf{x}(t) = e^{(\alpha+i\beta)t}(\mathbf{u} + i\mathbf{w}) = e^{\alpha t}[(\mathbf{u} + i\mathbf{w})(\cos(\beta t) + i\sin(\beta t))] \tag{23}$$

Expanding and taking real and imaginary parts:

$$\mathbf{x}_1(t) = e^{\alpha t}[\mathbf{u}\cos(\beta t) - \mathbf{w}\sin(\beta t)] \tag{24}$$

$$\mathbf{x}_2(t) = e^{\alpha t}[\mathbf{w}\cos(\beta t) + \mathbf{u}\sin(\beta t)] \tag{25}$$

These are two linearly independent real solutions.

Example 4.1: System with Complex Eigenvalues

onsider:

$$\mathbf{A} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \tag{26}$$

The characteristic equation is:

$$\det(\mathbf{A} - \lambda\mathbf{I}) = \lambda^2 + 1 = 0 \tag{27}$$

Eigenvalues: $\lambda = \pm i$
For $\lambda = i$:

$$(\mathbf{A} - i\mathbf{I})\mathbf{v} = \begin{pmatrix} -i & -1 \\ 1 & -i \end{pmatrix}\mathbf{v} = \mathbf{0} \tag{28}$$

This gives $\mathbf{v} = \begin{pmatrix} 1 \\ i \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + i\begin{pmatrix} 0 \\ 1 \end{pmatrix}$
So $\mathbf{u} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\mathbf{w} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.
With $\alpha = 0$ and $\beta = 1$, the real solutions are:

$$\mathbf{x}_1(t) = \begin{pmatrix} \cos t \\ \sin t \end{pmatrix} \tag{29}$$

$$\mathbf{x}_2(t) = \begin{pmatrix} -\sin t \\ \cos t \end{pmatrix} \tag{30}$$

General solution:

$$\mathbf{x}(t) = c_1\begin{pmatrix} \cos t \\ \sin t \end{pmatrix} + c_2\begin{pmatrix} -\sin t \\ \cos t \end{pmatrix} \tag{31}$$

5 Repeated Eigenvalues

When an eigenvalue has algebraic multiplicity greater than its geometric multiplicity, the matrix is not diagonalizable. In this case, we need generalized eigenvectors.

5.1 Geometric vs. Algebraic Multiplicity

Multiplicities For an eigenvalue λ:

Algebraic multiplicity the multiplicity of λ as a root of the characteristic polynomial.

Geometric multiplicity the dimension of the eigenspace, i.e., $\dim(\text{null}(\mathbf{A} - \lambda\mathbf{I}))$.

The geometric multiplicity is always less than or equal to the algebraic multiplicity.

5.2 Generalized Eigenvectors

When the geometric multiplicity is less than the algebraic multiplicity, we find generalized eigenvectors.

Generalized Eigenvectors For an eigenvalue λ with algebraic multiplicity m, the generalized eigenvectors of rank k satisfy:

$$(\mathbf{A} - \lambda\mathbf{I})^k\mathbf{v} = \mathbf{0} \tag{32}$$

The ordinary eigenvectors are generalized eigenvectors of rank 1.

5.3 Jordan Canonical Form

When a matrix is not diagonalizable, it can be transformed to Jordan canonical form:

$$\mathbf{A} = \mathbf{P}\mathbf{J}\mathbf{P}^{-1} \tag{33}$$

where $\mathbf{J}$ is a block diagonal matrix with Jordan blocks:

$$\mathbf{J}_k(\lambda) = \begin{pmatrix} \lambda & 1 & 0 & \cdots & 0 \\ 0 & \lambda & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda & 1 \\ 0 & 0 & \cdots & 0 & \lambda \end{pmatrix} \tag{34}$$

Example 5.1: Repeated Eigenvalue Case

onsider:

$$\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix} \tag{35}$$

The characteristic equation is:

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (2 - \lambda)^2 = 0 \tag{36}$$

So $\lambda = 2$ with algebraic multiplicity 2.
For the eigenspace:

$$(\mathbf{A} - 2\mathbf{I})\mathbf{v} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\mathbf{v} = \mathbf{0} \tag{37}$$

This gives only one linearly independent eigenvector: $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$.

For the generalized eigenvector, we solve:

$$(\mathbf{A} - 2\mathbf{I})\mathbf{v}_2 = \mathbf{v}_1 \tag{38}$$

This gives $\mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.
The solutions are:

$$\mathbf{x}_1(t) = e^{2t}\mathbf{v}_1 = e^{2t}\begin{pmatrix} 1 \\ 0 \end{pmatrix} \tag{39}$$

$$\mathbf{x}_2(t) = e^{2t}(t\mathbf{v}_1 + \mathbf{v}_2) = e^{2t}\begin{pmatrix} t \\ 1 \end{pmatrix} \tag{40}$$

6 Applications to Mechanical Systems

Eigenvalue methods are particularly powerful for analyzing mechanical systems with multiple degrees of freedom.

6.1 Normal Modes of Vibration

Consider a system of n masses connected by springs. The equations of motion are:

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{K}\mathbf{q} = \mathbf{0} \tag{41}$$

where $\mathbf{M}$ is the mass matrix and $\mathbf{K}$ is the stiffness matrix.
Assuming solutions of the form $\mathbf{q}(t) = \mathbf{v}e^{i\omega t}$:

$$(-\omega^2\mathbf{M} + \mathbf{K})\mathbf{v} = \mathbf{0} \tag{42}$$

This is a generalized eigenvalue problem:

$$\mathbf{K}\mathbf{v} = \omega^2\mathbf{M}\mathbf{v} \tag{43}$$

The eigenvalues ω_i^2 are the squares of the natural frequencies, and the eigenvectors $\mathbf{v}_i$ are the mode shapes.

6.2 Modal Analysis

The general solution is a superposition of normal modes:

$$\mathbf{q}(t) = \sum_{i=1}^{n}(A_i\cos(\omega_i t) + B_i\sin(\omega_i t))\mathbf{v}_i \tag{44}$$

Each mode oscillates independently at its natural frequency. This decomposition is fundamental to understanding vibrations in mechanical systems.

7 Stability Analysis via Eigenvalues

The eigenvalues of the coefficient matrix completely determine the stability of linear systems.

Theorem 7.1: Stability Criteria

or the linear system $\frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x}$:
Asymptotically Stable: All eigenvalues have negative real parts

$$\mathrm{Re}(\lambda_i) < 0 \text{ for all } i \tag{45}$$

Marginally Stable: All eigenvalues have non-positive real parts, with simple eigenvalues on the imaginary axis

$$\mathrm{Re}(\lambda_i) \leq 0 \text{ for all } i, \text{ with simple eigenvalues when } \mathrm{Re}(\lambda_i) = 0 \tag{46}$$

Unstable: At least one eigenvalue has positive real part

$$\mathrm{Re}(\lambda_i) > 0 \text{ for some } i \tag{47}$$

7.1 Routh-Hurwitz Criteria

For determining stability without explicitly computing eigenvalues, the Routh-Hurwitz criteria provide necessary and sufficient conditions based on the coefficients of the characteristic polynomial.

For a polynomial $p(\lambda) = a_n\lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_1\lambda + a_0$ with $a_n > 0$, all roots have negative real parts if and only if all the Hurwitz determinants are positive:

$$H_1 = a_{n-1}, \quad H_2 = \begin{vmatrix} a_{n-1} & a_{n-3} \\ a_n & a_{n-2} \end{vmatrix}, \quad H_3 = \begin{vmatrix} a_{n-1} & a_{n-3} & a_{n-5} \\ a_n & a_{n-2} & a_{n-4} \\ 0 & a_{n-1} & a_{n-3} \end{vmatrix}, \ldots \tag{48}$$

8 Advanced Topics

8.1 Matrix Functions

Beyond the matrix exponential, other matrix functions arise in applications:

Matrix Sine and Cosine:

$$\sin(\mathbf{A}t) = \sum_{k=0}^{\infty} \frac{(-1)^k (\mathbf{A}t)^{2k+1}}{(2k+1)!} \tag{49}$$

$$\cos(\mathbf{A}t) = \sum_{k=0}^{\infty} \frac{(-1)^k (\mathbf{A}t)^{2k}}{(2k)!} \tag{50}$$

Matrix Square Root: $\mathbf{A}^{1/2}$ such that $(\mathbf{A}^{1/2})^2 = \mathbf{A}$
Matrix Logarithm: $\log(\mathbf{A})$ such that $e^{\log(\mathbf{A})} = \mathbf{A}$

8.2 Pseudospectra

For non-normal matrices, eigenvalues can be highly sensitive to perturbations. Pseudospectra provide a more robust analysis tool:

$$\sigma_\epsilon(\mathbf{A}) = \{\lambda \in \mathbb{C} : \|(\lambda\mathbf{I} - \mathbf{A})^{-1}\| \geq 1/\epsilon\} \tag{51}$$

This set includes all points that are eigenvalues of some matrix within distance ϵ of $\mathbf{A}$.

Exercises

Exercise 1

Eigen decomposition and diagonalization.

1. For $A = \begin{pmatrix} 3 & 1 \\ 2 & 2 \end{pmatrix}$, find eigenvalues, eigenvectors, and an invertible P with $P^{-1}AP = D$.
2. Compute $e^{At} = Pe^{Dt}P^{-1}$ and solve $x' = Ax$, $x(0) = \binom{1}{0}$.
3. Classify the equilibrium at the origin (type and stability).

Exercise 2

Complex eigenvalues and real solutions.

1. For $A = \begin{pmatrix} \alpha & -\beta \\ \beta & \alpha \end{pmatrix}$ with $\beta \neq 0$, compute the eigenvalues and a complex eigenvector.
2. Produce two real, linearly independent solutions and write the general real solution of $x' = Ax$.
3. Describe trajectories in the plane for $\alpha < 0$, $\alpha = 0$, and $\alpha > 0$.

Exercise 3

Repeated eigenvalue and Jordan form.

1. For $A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$, find an eigenvector v_1 and a generalized eigenvector v_2 with $(A-2I)v_2 = v_1$.
2. Compute e^{At} explicitly and solve $x' = Ax$, $x(0) = \binom{0}{1}$.
3. State the equilibrium type and stability of the origin.

Exercise 4

Normal modes (two degrees of freedom).

1. Consider $q'' + Kq = 0$ with $K = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$. Find the natural frequencies and normalized mode shapes.
2. Write the general solution as a superposition of normal modes.
3. For $q(0) = \binom{1}{0}$ and $q'(0) = \binom{0}{0}$, determine the modal amplitudes.

Exercise 5

Stability via eigenvalues and Routh–Hurwitz.

1. For $A = \begin{pmatrix} -1 & -3 \\ 2 & -2 \end{pmatrix}$, classify stability from $\operatorname{tr} A$ and $\det A$, and confirm using eigenvalues.
2. Decide asymptotic stability of the polynomial $p(\lambda) = \lambda^3 + 3\lambda^2 + 2\lambda + 1$ using Routh–Hurwitz.

Exercise 6

Nonhomogeneous linear system.

1. Solve $x' = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} x + \binom{e^t}{\cos t}$ using variation of parameters (or undetermined coefficients when applicable).
2. Give the general solution and one solution satisfying $x(0) = \binom{0}{0}$.